Lecs 1 to 3

Statistics Formula Sheet — Detailed Edition

UNIT 1 — Classification and Tabulation of Data

When you have a huge list of raw numbers (e.g., marks of 200 students), it's hard to make sense of them. So you group the data into **classes** (also called **class intervals**) — ranges like 0–10, 10–20, 20–30, etc. This unit is about how those groups are built and described.

1. Class Width

$$\text{Class Width} = U - L$$

Class width is simply how "wide" each group/bucket is. If a class is "20–30", its width is 10.

Symbol	Meaning
U	**Upper limit** — the highest number that belongs to that class interval (e.g., in "20–30", $U = 30$)
L	**Lower limit** — the lowest number that belongs to that class interval (e.g., in "20–30", $L = 20$)

2. Class Mark (Midpoint)

$$x = \frac{L + U}{2}$$

Once data is grouped into a class like "20–30", you lose the exact individual values — you only know they fall somewhere in that range. So for calculations (like the mean), we assume every value in that class is represented by the **midpoint** of the interval. That midpoint is called the **class mark**.

Symbol	Meaning
x	Class mark — the midpoint value used to represent an entire class interval (e.g., for "20–30", $x = 25$)
L	Lower class limit (explained above)
U	Upper class limit (explained above)

UNIT 2 — Descriptive Statistics

"Descriptive statistics" just means numbers that *describe* a dataset in summary form — a single number telling you where the "center" of the data is, or how spread out it is — instead of having to look at every individual value.

A. Measures of Central Tendency

"Central tendency" means: **where is the middle / typical value of this dataset?** Mean, median, and mode are the three main ways to answer that.

1. Arithmetic Mean (Individual/Raw Data)

$$\bar{x} = \frac{\sum x}{n}$$

This is the everyday "average" — add up all the numbers and divide by how many numbers there are. Used when you have the raw, individual data points (not yet grouped into classes).

Symbol	Meaning
$\bar{x}$	(read "x-bar") — the **sample mean**, i.e., the average of the data you've collected
x	Each individual data value in your dataset (e.g., one student's marks)
$\sum$	The **summation symbol** — it just means "add up everything that comes after it." So $\sum x$ means "add up all the x values"
n	The **sample size** — the total count of data points you have (e.g., if you have 50 students' marks, $n = 50$)

2. Arithmetic Mean (Frequency Distribution)

$$\bar{x} = \frac{\sum fx}{\sum f}$$

When data is already grouped into classes with a count of how many items fall in each class, you use this version instead.

Symbol	Meaning
f	**Frequency** — this means "how many times something occurs" or "how many data points fall into this group." For example, if 12 students scored between 20–30, the frequency of that class is 12

Symbol	Meaning
x	The class mark (midpoint) representing that group of values, or the observation itself if ungrouped
fx	Frequency multiplied by the value — this weights each class mark by how many data points are actually in it
$\sum fx$	Add up the (frequency × value) for every single class
$\sum f$	Add up all the frequencies — this just gives you back the **total number of data points**, n

3. Assumed Mean Method

$$\bar{x} = A + \frac{\sum fd}{\sum f}, \qquad d = x - A$$

This is a shortcut calculation method — instead of multiplying large class-mark numbers directly, you pick a convenient reference point (the "assumed mean") and just work with small deviations from it, which is faster to calculate by hand.

Symbol	Meaning
A	**Assumed mean** — a value you guess or pick (usually the class mark of the class with the highest frequency, or one near the middle) purely to make arithmetic easier. It is *not* the actual mean — it's a reference point
d	**Deviation** — how far a class mark is from the assumed mean (just subtraction: $x - A$)
x, f	Same as defined above (class mark, frequency)

4. Step Deviation Method

$$\bar{x} = A + h\left(\frac{\sum fu}{\sum f}\right), \qquad u = \frac{x - A}{h}$$

An extension of the assumed mean method — here, the deviations are also divided by the class width, making the numbers you work with even smaller and easier to calculate with (especially useful when class width is the same for every class).

Symbol	Meaning
h	**Class width** (defined in Unit 1) — the size of each class interval, assumed constant across all classes for this method to work
u	**Step deviation** — the deviation d divided by the class width h; this scales the numbers down further

Symbol	Meaning
A, x, f	Same as above

5. Weighted Mean

$$\bar{x}_w = \frac{\sum wx}{\sum w}$$

A regular average treats every value equally. But sometimes some values matter more than others (e.g., a final exam counts for more than a quiz). The weighted mean lets you assign different levels of importance to each value.

Symbol	Meaning
$\bar{x}_w$	The **weighted mean** — the average after accounting for different importance levels
w	**Weight** — a number representing how much importance/emphasis a particular value should get (e.g., exam weight = 70%, quiz weight = 30%)
x	The value/observation being weighted

6. Median (Ungrouped Data)

The **median** is the value that sits exactly in the middle when all data is arranged in order (smallest to largest) — half the data is below it, half is above it. Unlike the mean, it isn't affected much by extremely large or small outlier values.

Odd number of observations:

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{observation}$$

Even number of observations (there's no single middle number, so you average the two middle ones):

$$\text{Median} = \frac{x_{n/2} + x_{(n/2)+1}}{2}$$

Symbol	Meaning
n	Total number of observations (data points)
x_i	The value that sits in the i^{th} position, **after you've sorted the data from smallest to largest**

7. Median (Grouped Data)

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) h$$

When data is grouped into classes, you can't just "pick the middle value" directly since individual values are hidden inside ranges. This formula estimates where the median falls within the correct class.

Symbol	Meaning
L	Lower boundary of the **median class** — the specific class interval that contains the middle of the data
N	Total number of observations (sum of all frequencies)
cf	**Cumulative frequency** — running total of frequencies *up to but not including* the median class. (Cumulative = "added up as you go." E.g., if classes have frequencies 5, 8, 12, the cumulative frequency after the 2nd class is 5+8=13)
f	Frequency of the median class specifically
h	Class width

8. Mode (Grouped Data)

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) h$$

The **mode** is the value or class that occurs most frequently — the "most popular" value in the dataset.

Symbol	Meaning
L	Lower boundary of the **modal class** — the class interval with the highest frequency of all
f_1	Frequency of the modal class (the highest frequency)
f_0	Frequency of the class that comes right **before** the modal class
f_2	Frequency of the class that comes right **after** the modal class
h	Class width

9. Empirical Relation

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

An approximate shortcut relationship between the three central-tendency measures — useful when you want a rough estimate of the mode without doing the full modal-class calculation, based only on the mean and median (this relation is only an approximation, valid for moderately skewed distributions).

B. Quartiles

Quartiles divide your ordered data into 4 equal parts (like the median divides it into 2 parts). Each quarter contains 25% of the data.

First & Third Quartile

$$Q_1 = L + \left(\frac{\frac{N}{4} - cf}{f} \right) h \qquad Q_3 = L + \left(\frac{\frac{3N}{4} - cf}{f} \right) h$$

Symbol	Meaning
Q_1	**First quartile** — the value below which the lowest 25% of the data falls (also called the 25th percentile)
Q_3	**Third quartile** — the value below which the lowest 75% of the data falls (75th percentile)
L	Lower boundary of the class that contains Q_1 or Q_3 (found the same way as the median class, just using $N/4$ or $3N/4$ instead of $N/2$)
N	Total frequency (total number of data points)
cf	Cumulative frequency before that particular quartile's class
f	Frequency of the class containing the quartile
h	Class width

(Note: Q_2, the second quartile, is just the median — it splits data at the 50% mark.)

Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$

This tells you the spread of the **middle 50%** of your data — i.e., ignoring the bottom 25% and top 25% (which are more likely to contain outliers), how wide is the "core" of the data?

Quartile Deviation (QD)

$$QD = \frac{Q_3 - Q_1}{2}$$

Also called the **semi-interquartile range** — literally half the IQR. It's a measure of spread that, like the median, isn't thrown off by extreme outlier values.

Coefficient of Quartile Deviation

$$\frac{Q_3 - Q_1}{Q_3 + Q_1}$$

This turns quartile deviation into a **relative** (ratio) measure with no units attached, so you can fairly compare the "spread" of two datasets even if they're measured in different units (e.g., comparing spread of heights in cm vs. weights in kg).

C. Distribution Visualization (Graphs)

Relative Frequency

$$\text{Relative Frequency} = \frac{f}{N}$$

This converts a raw count (frequency) into a **proportion** (fraction of the whole), so you can see what share of the total each class represents.

Symbol	Meaning
f	Frequency of a specific class (how many data points fall into it)
N	Total frequency (total number of data points across all classes combined)

Percentage Frequency

$$\text{Percentage Frequency} = \frac{f}{N} \times 100$$

Same idea as relative frequency, just multiplied by 100 to express it as a percentage instead of a decimal fraction.

Pie Chart Angle

$$\theta = \frac{\text{Category Frequency}}{\text{Total Frequency}} \times 360^\circ$$

A full circle (pie chart) spans 360°. To show a category's share visually as a "slice" of the pie, you find what fraction of the total that category represents, then convert that fraction into degrees out of 360.

Symbol	Meaning
θ	(read "theta") — the angle, in degrees, that a category's slice should occupy in the pie chart
Category Frequency	How many data points fall into that one specific category
Total Frequency	Sum of frequencies across *all* categories

UNIT 3 — Measures of Variability

While Unit 2 was about "where is the center of the data," this unit is about "**how spread out** is the data around that center?" Two datasets can have the exact same mean but be wildly different in how scattered their values are — variability measures capture that difference.

A. Dispersion, Skewness and Kurtosis

1. Range

$$R = L - S$$

The simplest possible measure of spread — just the gap between the biggest and smallest value.

Symbol	Meaning
L	The **largest** (maximum) value in the entire dataset
S	The **smallest** (minimum) value in the entire dataset

2. Coefficient of Range

$$\frac{L - S}{L + S}$$

The range expressed as a ratio rather than a raw number, so it can be compared across datasets of different scales/units.

3. Mean Absolute Deviation (MAD)

$$MAD = \frac{\sum |x - \bar{x}|}{n}$$

For every data point, find how far it is from the mean (ignoring whether it's above or below — that's what the | | "absolute value" bars mean, they just make everything positive). Then average all those distances. This tells you, on average, how far a typical data point sits from the mean.

Symbol	Meaning
x	Each individual data value
$\bar{x}$	The mean
$	x - \bar{x}
n	Number of observations

4. Population Variance

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$$

Variance also measures spread, but instead of taking the absolute value of each deviation, it **squares** each deviation (which also removes negative signs, but additionally gives extra weight to larger deviations — meaning outliers affect variance more heavily than MAD). This is calculated over an entire **population** — meaning literally every single member of the group you care about (not just a smaller sample from it).

Symbol	Meaning
σ^2	(read "sigma squared") — **population variance**
x	Each value in the population
μ	(read "mu") — the **population mean** (the true average of the entire population, not just a sample)
N	**Population size** — the count of every single member in the entire group being studied

5. Sample Variance

$$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$$

In real life, you usually can't measure an entire population (e.g., every voter in a country) — instead you take a smaller **sample** and use it to estimate the population's characteristics. This is the sample version of variance.

Symbol	Meaning
s^2	**Sample variance** — variance calculated from a sample, used to *estimate* the true population variance
$\bar{x}$	Sample mean
$n-1$	Instead of dividing by n (sample size), we divide by $n-1$. This is called **degrees of freedom**, and the "−1" is known as **Bessel's correction** — it corrects for the fact that a sample tends to slightly underestimate the true population variance, so dividing by a slightly smaller number nudges the estimate to be more accurate

6. Standard Deviation

$$\sigma = \sqrt{\sigma^2} \qquad\qquad s = \sqrt{s^2}$$

Variance is in **squared units** (e.g., if your data is in rupees, variance is in "rupees²," which is hard to interpret intuitively). Standard deviation just takes the square root of variance, bringing it back to the original, easy-to-understand units. This is the most commonly used measure of spread.

Symbol	Meaning
σ	Population standard deviation
s	Sample standard deviation

7. Coefficient of Variation (CV)

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

Standard deviation alone doesn't tell you if the spread is "a lot" or "a little" — that depends on the scale of the data. CV expresses standard deviation as a **percentage of the mean**, letting you compare variability between two totally different datasets (e.g., comparing variability of stock prices in ₹ vs. variability of returns in %).

8. Chebyshev's Theorem

$$\text{Minimum proportion} = 1 - \frac{1}{k^2} \quad \text{within } \mu \pm k\sigma$$

This is a general rule (works for *any* shape of data distribution, not just the bell-curve/normal one) that tells you the minimum guaranteed percentage of your data that will fall within a certain number of standard deviations from the mean.

Symbol	Meaning
k	How many standard deviations away from the mean you're looking at (must be greater than 1). E.g., $k = 2$ asks "what % of data is guaranteed to fall within 2 standard deviations of the mean?"
$\mu \pm k\sigma$	The range from (μ − k standard deviations) to (μ + k standard deviations)

9. Pearson's Coefficient of Skewness

Using Mode:

$$Sk = \frac{\bar{x} - \text{Mode}}{\sigma}$$

Using Median:

$$Sk = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

Skewness measures whether your data is symmetric or lopsided. If the mean is much bigger than the median/mode, the data has a long tail stretching to the right (**positive/right skew**). If the mean is much smaller, the tail stretches left (**negative/left skew**). A value near 0 means the data is roughly symmetric.

10. Kurtosis

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \qquad \gamma_2 = \beta_2 - 3$$

Kurtosis describes the "shape" of the distribution's peak and tails — specifically, whether it has a sharp, tall peak with heavy tails (more extreme outliers than normal), or a flatter peak with thin tails.

Symbol	Meaning
β_2	Kurtosis coefficient
γ_2	**Excess kurtosis** — kurtosis measured relative to the normal (bell-curve) distribution, which itself has a β_2 value of exactly 3. So $\gamma_2 > 0$ means "more extreme/peaked than normal," and $\gamma_2 < 0$ means "flatter than normal"
μ_2	The **second central moment** — this is just another name for variance (average of squared deviations from the mean)
μ_4	The **fourth central moment** — average of deviations from the mean raised to the 4th power; this exaggerates the effect of extreme/outlier values even more than variance does, which is what makes it useful for detecting heavy tails

B. Correlations and Visualizations

While the measures above described *one* variable at a time, correlation describes the **relationship between two variables** — does one tend to increase when the other increases (or decreases)?

1. Sample Covariance

Population:

$$Cov(X,Y) = \frac{\sum(X-\mu_X)(Y-\mu_Y)}{N}$$

Sample:

$$Cov(X,Y) = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{n-1}$$

Covariance tells you the **direction** in which two variables move together. If, when X is above its mean, Y also tends to be above its mean (and vice versa), covariance is positive. If one tends to be above its mean while the other is below, covariance is negative.

Symbol	Meaning
X, Y	The two variables you're comparing (e.g., X = hours studied, Y = exam score)
μ_X, μ_Y	Population means of X and Y respectively
$\bar{X}, \bar{Y}$	Sample means of X and Y respectively
N	Population size
n	Sample size

Limitation: covariance's actual numeric size depends on the units of X and Y, which makes it hard to interpret "how strong" the relationship is on its own — that's what correlation (below) fixes.

2. Pearson's Correlation Coefficient

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left(n\sum x^2 - (\sum x)^2\right)\left(n\sum y^2 - (\sum y)^2\right)}}$$

This is covariance, but standardized (rescaled) so that it always falls between −1 and +1, regardless of the units X and Y are measured in. This makes it easy to judge both the **direction** and the **strength** of a straight-line (linear) relationship between two variables.

Symbol	Meaning
r	Pearson's correlation coefficient
x, y	A pair of values — one from each variable, measured on the same subject/observation (e.g., one student's study-hours and exam-score)
n	Number of paired observations

Range: $-1 \leq r \leq 1$

- $r = +1$: perfect positive relationship (as one goes up, the other always goes up proportionally)
- $r = -1$: perfect negative relationship (as one goes up, the other always goes down proportionally)
- $r = 0$: no linear relationship at all

3. Spearman's Rank Correlation

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Instead of using the raw numeric values like Pearson's does, Spearman's correlation first converts each variable's values into **ranks** (1st, 2nd, 3rd, $\ldots$) and then measures correlation between the ranks. This makes it useful when the relationship between two variables is consistently increasing/decreasing but not necessarily in a straight line, or when you only have ranked (ordinal) data to begin with.

Symbol	Meaning
ρ	(read "rho") — Spearman's rank correlation coefficient
d	The difference between the ranks of a pair of observations (e.g., a student ranks 3rd in study-hours and 5th in exam-score — $d = 3 - 5 = -2$)
n	Number of paired observations

Quick Reference — Common Symbols (Plain-English Glossary)

Symbol	Meaning
$\sum$	"Sum of" — add up every value that follows this symbol
$\bar{x}$	Sample mean (average of a sample)

Symbol	Meaning
μ	Population mean (average of an entire population)
σ	Population standard deviation (typical distance of a data point from the population mean, in original units)
s	Sample standard deviation (same idea, but estimated from a sample)
σ^2	Population variance (standard deviation, before taking the square root — in squared units)
s^2	Sample variance
n	Sample size — how many data points are in your sample
N	Population size, or total frequency — the full count of everything (context-dependent)
f	Frequency — how many times a value, or how many data points, occur in a given class/category
cf	Cumulative frequency — the running total of frequencies as you move through the classes, from the first class up to and including a given point
h	Class width — the size (span) of each class interval
Q_1, Q_2, Q_3	First, second (= median), and third quartile — the values splitting ordered data into four equal 25% chunks
R	Range — biggest value minus smallest value
CV	Coefficient of Variation — standard deviation as a % of the mean
$Cov(X, Y)$	Covariance — direction of the relationship between two variables
r	Pearson correlation coefficient — strength & direction of a straight-line relationship between two variables (−1 to +1)
ρ	Spearman rank correlation coefficient — same idea as r, but based on ranks instead of raw values